A SURJECTIVE NON-IDENTITY SELFMAP OF A HAUSDORFF SPACE IS COVERED BY A PROPER CLOSED SUBSET AND ITS IMAGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Hausdorff separation axiom and the notion of a proper closed subset. Set up in the author's section on extremally disconnected spaces, where the counted statement occurs. Supplies the disjoint open neighbourhoods used to define the closed subset.

Lemma 0.1. *Let $f : X \to X$ be a surjective continuous selfmap of a Hausdorff topological space. If f is not id_X, then there exists a proper closed subset $E \subset X$ such that $X = E \cup f(E)$.*

Proof. Pick $p \in X$ with $f(p) \neq p$. Choose disjoint open neighbourhoods $p \in U$, $f(p) \in V$ and set $E = X \setminus U \cap f^{-1}(V)$. Then $p \notin E$ hence E is a proper closed subset. If $x \in X$, then either $x \in E$, or if not, then $x \in U \cap f^{-1}(V)$. Writing $x = f(y)$ (possible as f is surjective). If $y \in U \cap f^{-1}(V)$ then we would have $x = f(y) \in V$ which is a contradiction with $x \in U$. Hence $y \in E$ and $x \in f(E)$. □

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